

Project Report: Project Zenith

Real-Time Orbital Mechanics & Visualization

1. Executive Summary

Project Zenith is a high-fidelity, interactive space-tracking platform designed to provide real-time planetary mechanics, artificial satellite monitoring, and cinematic 3D rendering. Developed for the **SRMIST AARUUSH '26 National Technical Championship**, this project integrates live telemetry with a high-performance 3D engine to deliver an immersive astronomical experience.

- Status:** Qualified for Round 2 Development Sprint
- Submission Score:** 98.40/100
- Developer:** AstralWeb Innovate

2. Platform Overview

Project Zenith serves as an advanced astronomical dashboard. Key capabilities include:

- Tracking of real-time orbital trajectories.
- Satellite position monitoring via live data streams.
- Orthographic radar system for precise geodetic observation.

3. Technical Architecture

The platform utilizes a modern, high-performance web stack:

- Core Framework:** [React](#) with [TypeScript](#) for robust, type-safe architecture.
- Styling:** [Tailwind CSS v4](#) for optimized UI responsiveness.
- Build Environment:** [Vite](#) for high-speed development and bundling.
- AI Integration:** [Gemini API](#) for cognitive celestial reasoning and advanced stargazing analysis.

4. Data Strategy

Accuracy is maintained through the following active data pipelines:

- NASA Horizons:** Utilized for planetary ephemeris and precise celestial position vectors.
- OpenNotify:** Integrated for live International Space Station (ISS) tracking and low-orbit altitude mapping.
- CelesTrak:** Used for processing Two-Line Element (TLE) datasets for over 6,000 active satellites.

6. Celestial Visibility Logic

The engine determines the visibility of astronomical objects using the following mathematical model:

- Variables:**
 - α = Azimuth (Observer, Object)

- ϵ = Elevation (Observer, Object)
- Visibility Condition:** Visible only if $\epsilon > 0^\circ$ (above the horizon).
- Viability Index:** Calculated as:

$$\text{Viability Index} = (\text{Seeing Value} \times 20) - \text{Cloud Cover} \%$$

6. Development Roadmap

Phase	Focus	Status
Week 1	Data documentation & TLE vector blueprints	Completed
Week 2	Radar coordinates plotter & UI mockups	Completed
Week 3-4	Live backend sync & AI analysis modules	In Progress

7. Contact Information

For inquiries regarding this project or the AstralWeb Innovate portfolio, please reach out via:

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- Live Project:** [Project Zenith Website](#)